



Louise LECLERC

PhD student in Computer Sciences and Mathematics at LIX
(Institut Polytechnique de Paris)

website : <https://luyise.github.io/>

1 Studies

- PhD (ongoing) : A Directed Homotopy Type Theory. under the supervision of Samuel Mimram, at Polytechnique Paris.
- Pre-thesis internship (4 months) under the supervision of Samuel Mimram, at Polytechnique. On the subject of directed type theory.
- Master's degree (Bac+5) : "Fundamental Mathematics" from Paris-Cité University. Graduated with highest honors (Mention Très Bien).
- Seven-month internship at IRIF (Paris), under the supervision of Pierre-Louis Curien. On the subject of higher categories. (Study of various definitions of opetope)
- Four-month internship under the supervision of Thierry Coquand, in the "Logic and Types" team at Chalmers University, in Gothenburg (Sweden), on the subject of Homotopy Type Theory.
- Short internship (1.5 months) at INRIA (Paris-Saclay) under the supervision of Sylvie Boldo. About the formalization of mathematics using proof assistants.
- Bachelor's degree (Bac+3) : mathematics degree from the École Normale Supérieure Paris
- Entrance exam to the ENS : 16th place in the 2020 exam session
- Bac+1/Bac+2 : CPGE MPSI (2018-2019) then MP* (2019-2020) at Lycée Descartes (Tours).
- High school : Baccalaureate in Scientific Engineering Sciences, specializing in Mathematics, European English section (2018) at the end of schooling at St-Denis International School (Loches).

2 Formation

- Mathematics :
 - Master's level mathematics, primarily :
 - * Logic : Intuitionistic Type Theory, Homotopy Type Theory. Basic knowledge of model theory and categorical logic.
 - * Algebra : Opetopes, Lie Theory, Category theory, group theory, group representations, commutative algebra.
 - * Homotopy : Model categories, standard homotopy theory.
 - * Geometry : differential and Riemannian geometry, general topology.
 - * Analysis : Real analysis, complex analysis and dynamical systems at M1 level.
 - Undergraduate mathematics.
- Computer science knowledge, equivalent to L2 level, mainly in functional programming.
 - Languages known : OCaml, Rocq, Agda, Processing, familiarity with Rust, Python and Haskell.
 - Experience with the proof assistants Rocq and Agda.
 - A third-year undergraduate level semantics course has been validated at the ENS.
- Knowledge of physics, equivalent to L2 level.
 - quantum mechanics, statistical physics, particle physics (L3 level courses validated at ENS-PSL)
 - Classical mechanics, non-relativistic electromagnetism, thermodynamics. (studied in CPGE)
- Languages known : French (native language), English (level B2).

3 Publications and Talks

- Presentation at the higher categories mini workshopp in Paris : "A type theory for cellular spaces" : [slides](#).
- Presentation at the HoTT-UF Workshopp 2025, in Genoa : "Towards a type theory for (∞, ω) —categories" : [slides](#).
- Presentation at LHC 2024 : "Different approaches to opetopes".
- Papers written at the end of the internship led by Pierre-Louis Curien : "A poset-like approach to positive opetopes" : [page HAL](#)
and "Opetopes as poset and zoom complexes" : [page ArXiv](#)
- Paper written at the end of the internship led by Sylvie Boldo : "A Coq Formalization of the Bochner integral" : [page HAL](#)